Beliz Kaleli

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Principal Security Research Engineer / Team Lead building AI-powered security systems for large-scale real-world threat detection in high-stakes environments. Proven track record of delivering high-performance production-grade systems that operate at scale; vulnerability research (CVEs, IOCs), TTP analysis, browser exploit identification and malicious infrastructure analysis. Primary inventor on 3+ patents, author of 2 award-winning research and 10+ high-impact threat intelligence publications. Core expertise includes designing autonomous AI agents for detection, with a focus on emerging attack vectors such as indirect prompt injections, vulnerability scanning, GenAI-driven phishing and malware delivery methods.

Work Experience

Principal Security Research Engineer | *Palo Alto Networks, Santa Clara, CA, USA* Oct 2026 – Present

Sr. Staff Security Research Engineer | *Palo Alto Networks, Santa Clara, CA, USA* Oct 2024 – Oct 2026

- Led a cross-functional team of Security Research Engineers to design, prototype, and deploy innovative AI/ML systems for detecting phishing attacks at scale.
- Led a cross-functional effort to design and develop a hybrid, multi-network vulnerability scanning detection system combining volume-based features, threat signatures, and ML. Currently blocks ~1M malicious requests per day real time (patented; first-author paper and article).
- Built an AI-powered indirect prompt injection (IDPI) detection system protecting AI agents from web-based attacks; achieved first real-world detection of high-severity IDPI (AI reviewer bypass, data exfiltration, prompt leakage, and DoS) (patent pending; invited talk at Stanford University; first-author article and post).
- Built an autonomous AI agent crawler to solve captchas, authenticate, and navigate complex page interactions to access previously inaccessible content (gated malicious downloads, phishing forms) critical for detection (placed 2nd among 30 teams at the Google Cloud/Palo Alto Networks Agentic AI Hackathon).
- Led a cross-team effort to design (PRD) and develop an AI-powered system for detecting OAuth flow abuse in real-time production traffic (patent pending; published 3 posts).
- Designed and implemented an agentic brand impersonation detection system using LLMs with RAG, multimodal analysis, and integrated tools to identify phishing attacks against global and regional brands.
- Primary inventor on 3+ patents (7+ in total) for AI-based security detection techniques, covering phishing, scams, indirect prompt injection, vulnerability scanning, and domain squatting attacks.
- Detected 30+ novel attack campaigns (IDPI, OAuth abuse, browser-in-the-middle, GenAI-based phishing), documented in 10 high-impact threat intelligence publications (7 as first author).

Staff Security Research Engineer | *Palo Alto Networks, Santa Clara, CA, USA* May 2023 – Oct 2024

Security Researcher Intern | *Palo Alto Networks, Santa Clara, CA, USA* Jun 2022 – Nov 2022

Software Engineer | *AYESAS, Ankara, Turkey* Sep 2017 – Jun 2018

Skills

Languages: Python, Go, SQL, C, C++, HTML/CSS, JavaScript, Typescript, Java

Tools & Frameworks: GCP, BigQuery, Vertex AI, Claude Code, Google ADK, VirusTotal, SIEM, MCP, Git, CI/CD, Docker, Kubernetes, Pub/Sub, Terraform, REST, gRPC, MongoDB, Scikit-Learn, LangChain, AWS

Awards & Honors

2nd Place at Hackathon | *Google Cloud/Palo Alto Networks Agentic AI Hackathon* 2026

Best Student Paper Award | *Security and Privacy in Communication Networks (SecureComm)* 2024

Best Student Paper Award Finalist | *The Center for Information & Systems Engineering (CISE)* 2020

Best Paper Award | *Detection of Intrusions and Malware, and Vulnerability Assessment (DIMVA)* 2019

Patents

- A Deep Learning Approach for Real-time Scanning URLs Detection Using IUPG Framework (US-20260254819-A1), Sep 2026 — Patented
  *T. Li, B. Kutt, **B. Kaleli**, F. Liu, P. Peng, O. Starov, W. Hewlett*
- Agentic Web-Based Indirect Prompt Injection (IDPI) Detector — Filed, Apr 2026
  ***B. Kaleli**, S. Farooqi, N. Mohamed, O. Starov*
- Detecting Scanning and Attacking URLs in Network Traffic — Patented (US-12483573-B2), Nov 2025
  ***B. Kaleli**, F. Liu, O. Starov*
- Phantom Squatting: Anticipating GenAI-Hallucinated URLs Before Attackers Do — Filed, Oct 2025
  *K. Nagaraj, D. Marty, **B. Kaleli**, O. Starov*
- Detecting Defensively Registered Domains Using LLMs — Filed, Oct 2025
  *N. Mohamed, J. Chen, O. Starov, **B. Kaleli***
- AVA: Authentication Verification Analyzer for Detecting OAuth Flow Abuses — Filed, Jun 2025
  ***B. Kaleli**, D. Marty, L. Hu, O. Starov*
- A Grounded AI-based Pipeline for Detecting Social Engineering Attacks — Filed, Jun 2025
  *L. Hu, F. Liu, **B. Kaleli**, O. Starov*

Academic Papers

- ScanWars: (A Multi-network Approach to Detecting and Analyzing) The Rise of Scanning Activity — 2025
  ***B. Kaleli**, T. Li, F. Liu, O. Starov, M. Egele, G. Stringhini*
  Symposium on Electronic Crime Research (eCrime)
- EasyCSPeasy: A Server-side and Language-agnostic XSS Mitigation — 2024
  ***B. Kaleli**, M. Egele, G. Stringhini*
  *Security and Privacy in Communication Networks (SecureComm) – **Best Paper Award***
- To Err.Is Human: Characterizing the Threat of Unintended URLs in Social Media — 2021
  ***B. Kaleli**, B. Kondracki, M. Egele, G. Stringhini, N. Nikiforakis*
  Network and Distributed System Security Symposium (NDSS)
- Studying the Privacy Issues of the Incorrect Use of the Feature Policy — 2020
  ***B. Kaleli**, M. Egele, G. Stringhini*
  Workshop on Measurements, Attacks, and Defenses for the Web (MADWeb)
- On the Perils of Leaking Referrers in Online Collaboration Services — 2019
  ***B. Kaleli**, M. Egele, G. Stringhini*
  *International Conference on Detection of Intrusions and Malware, and Vulnerability Assessment (DIMVA) – **Best Paper Award***

Invited Talks & Service

Invited Speaker at Stanford University | Real World AI Security Conference — 2026
Program Committee Member | Annual Computer Security Applications Conference (ACSAC) — 2025
Invited Speaker at University of Michigan | Center for Computer Security and Society (SECRIT) — 2021

Other Publications

Articles

- The Current Landscape of OAuth Attacks — Oct 2026
  ***B. Kaleli**, D. Marty, O. Starov, S. Shuang*
- Fooling AI Agents: Web-Based Indirect Prompt Injection Observed in the Wild — Mar 2026
  ***B. Kaleli**, S. Farooqi, O. Starov, N. Mohamed*
- Fashionable Phishing Bait: GenAI on the Hook — Aug 2025
  *Z. You, **B. Kaleli**, L. Hu, W. Wang*
- It Was Not Me! Malware-Initiated Vulnerability Scanning Is on the Rise — Apr 2024
  ***B. Kaleli**, F. Liu, P. Peng, O. Starov, J. Allen, S. Springer*

Posts

- New Browser-in-the-Browser Phishing Campaign — Jun 2026
  ***B. Kaleli**, G. Rane, O. Starov*
- Indirect Prompt Injection to Push a Malicious Chrome Extension — May 2026
  ***B. Kaleli**, S. Farooqi, N. Mohamed, O. Starov*
- Phishing Campaign Pivots to Sophisticated Methods — May 2026
  ***B. Kaleli**, O. Starov*

- Real-World Example of Malicious Indirect Prompt Injection — Dec 2025
  ***B. Kaleli**, S. Farooqi, O. Starov*
- Crypto-themed Emails with Links to Sites Posing as Cryptocurrency Platforms — Oct 2025
  ***B. Kaleli**, O. Starov*
- Phishing Campaigns Impersonate Major Business Platforms and Abuse OAuth Flows — Oct 2025
  ***B. Kaleli**, O. Starov*
- Ongoing Phishing Campaign Using Browser-In-The-Middle Technique — Sep 2025
  ***B. Kaleli**, O. Starov, L. Hu, S. Shuang, N. Mohamed, B. Duncan*

Education

Boston University — *Boston, MA, USA*
Ph.D. & M.S., Computer Engineering, ECE | GPA: 3.8/4.0 — Jan 2025

Middle East Technical University — *Ankara, Turkey*
B.S., Electrical and Electronics Engineering — Sep 2017